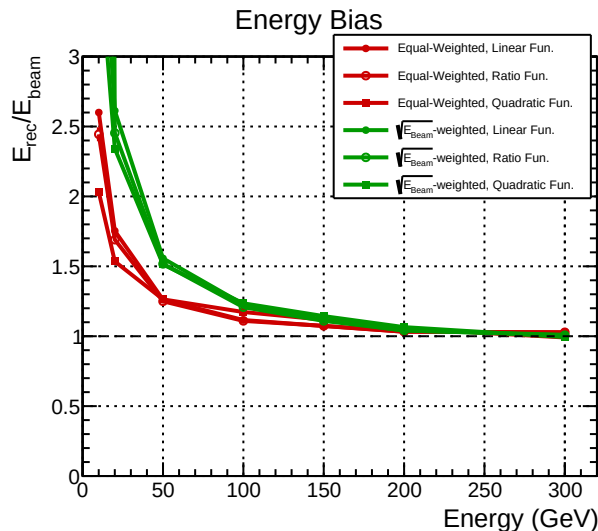
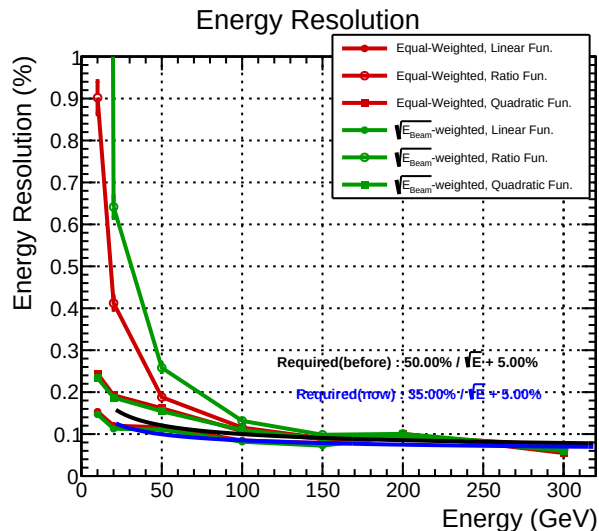


Neutron Beam Energy Regression



- Optimal Method: Sqrt(Ebeam) weighting combined with the Ratio Method yielded the best results.
- Energy Resolution: neutrons < 50 GeV does not meet the requirements.
- Energy Bias: A bias of approximately 2% is observed for neutron > 50GeV. Large bias for low energy region (neutron < 50GeV).
- The small fraction of neutron energy deposited in the scintillator tile (3%–6%) limits regression precision for beam energies under 50 GeV?